

Problem (1)

$$a-) \frac{dF_A}{dW} = -\eta k'' s_a P_c (1-E) C_A$$

$$\frac{dC_A}{dW} = \frac{-\eta k'' s_a P_c (1-E) C_A}{N} \quad (F_A = C_A V)$$

$$\int \frac{dC_A}{C_A} = - \int \left\{ \frac{\eta k'' s_a P_c (1-E)}{N} \right\} dW \quad \left\{ A = 0.00924 \right.$$

$$\ln(C_A/C_{A0}) = -A W (1-E) P_c$$

$$\ln(C_A/C_{A0}) = -0.00924 (1-E) P_c$$

$$b-) k_2 = k_1 \exp\left(E \left(\frac{1}{T_1} + \frac{1}{T_2} \right)\right) = 1 \times 10^{-5} \exp\left(100 \times \left(\frac{1}{400} + \frac{1}{420} \right)\right)$$

$$k_2 = 1.625 \times 10^{-5}$$

$$\frac{A_2}{A_1} = \frac{k_2'' \eta_2}{k_1'' \eta_1} = \frac{k_2 \sqrt{k_1}}{k_1 \sqrt{k_2}} = \frac{1.625 \times 10^{-5} \sqrt{1 \times 10^{-5}}}{1 \times 10^{-5} \sqrt{1.625 \times 10^{-5}}}$$

$$\frac{A_2}{A_1} = 1.274 \longrightarrow A_2 = 0.11716$$

$$\ln\left(\frac{C_A}{C_{A_0}}\right) = \ln\left(\frac{C_{A_0}(1-x)}{C_{A_0}}\right) = \ln(1-0.5) = \ln(0.5) = -4w(1-E)P_c$$

$$-0.6931 = -0.01178(1-0.4)(2.5)w$$

$$w = 39.227 \text{ g of catalyst.}$$

Problem (2) :-



$$-r'_{A_s} = k' C_{A_s} \quad Sh = 100 Re^{1/2}$$

$$Re = \frac{10 \text{ cm/s} \times 0.1 \text{ cm}}{0.02 \text{ cm}^2/\text{s}} = 50$$

$$Sh = 100 \sqrt{50} = 707.107$$

$$k_c = \frac{707.107}{10} = 70.711$$

$$k'' = k' / s_q = 0.01 / 60 = 1.67 \times 10^{-4}$$

$$k_{eff} = k_c k'' / (k_c + k'') = (70.71)(1.67 \times 10^{-4}) / (70.71 + 1.67 \times 10^{-4})$$

$$k_{eff} = 1.67 \times 10^{-4} \text{ cm/s}$$

$$\frac{dF_A}{dw} = k_{eff} C_A \text{ area} \rightarrow \ln \frac{C_A}{C_{A_0}} = \frac{-w}{N} \quad k_{eff} \times \text{area}$$

$$w = \ln(0.4) / (-10^4 \times 1.67 \times 10^{-4} \times 60) = 0.00914 \text{ g}$$